Find the Error

Grade 7 · 7.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Mia rides $\frac{1}{2}$ mile in $\frac{1}{4}$ hour. What is her unit rate in miles per hour? Student's answer: $\frac{1}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied $\frac{1}{2}$ by $\frac{1}{4}$. Divide: $\frac{1}{2} \div \frac{1}{4}$ = 2 miles per hour.**

Correct answer: **2 miles per hour**

Find the Error

Grade 7 · 7.RP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$\frac{3}{4}$ pound of cheese costs $6. What is the price per pound? Student's answer: $4.50

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied $6 by $\frac{3}{4}$. Divide: $6 ÷ $\frac{3}{4}$ = $8 per pound.**

Correct answer: **$8 per pound**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Mario walks $1\frac{1}{2}$ miles in $\frac{1}{3}$ hour. What is his speed in miles per hour? Student's answer: $\frac{1}{2}$ mile per hour

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student divided the wrong way. $1\frac{1}{2} \div \frac{1}{3} = 4\frac{1}{2}$ miles per hour.**

Correct answer: **$4\frac{1}{2}$ miles per hour**

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is this table proportional? x = 1, 2, 3, 4 and y = 3, 6, 9, 10. Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **For (4, 10), y/x = 2.5, but the other pairs give 3. The table is not proportional.**

Correct answer: **No**

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

4 items cost $12 in total. What is the constant of proportionality for cost per item? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the number of items. The constant is $12 ÷ 4 = $3 per item.**

Correct answer: 3

Find the Error

Grade 7 · 7.RP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The equation is y = 2.5x. What is y when x = 8? Student's answer: 10.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2.5 and 8. Multiply: 2.5 × 8 = 20.**

Correct answer: **20**

Find the Error

Grade 7 · 7.RP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A restaurant bill is $40. What is a 15% tip? Student's answer: $15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found 37.5%, not 15%. 0.15 × $40 = $6.**

Correct answer: $6

Find the Error

Grade 7 · 7.RP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An $80 item increases in price by 25%. What is the new price? Student's answer: $105

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 25% of the wrong amount. 25% of $80 is $20, so the new price is $100.**

Correct answer: **$100**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A $25 book has an 8% sales tax. What is the total price? Student's answer: $2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$2 is the tax alone. Add it to the price: $25 + $2 = $27.**

Correct answer: **$27**

Find the Error

Grade 7 · 7.NS.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add -3 + 8. Student's answer: -11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the absolute values. 8 is larger and positive, so -3 + 8 = 5.**

Correct answer: 5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Subtract 4 - (-6). Student's answer: -2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 6. Subtracting -6 is the same as adding 6: 4 + 6 = 10.**

Correct answer: 10

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add -2.5 + (-4). Student's answer: 6.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The sum of two negatives is negative. -2.5 + (-4) = -6.5.**

Correct answer: -6.5

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply (-4) × (-6). Student's answer: -24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Two negatives multiply to a positive. (-4) × (-6) = 24.**

Correct answer: **24**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Divide (-12) ÷ 3. Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A negative divided by a positive is negative. -12 ÷ 3 = -4.**

Correct answer: **-4**

Find the Error

Grade 7 · 7.NS.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(-\frac{2}{3}) \times (\frac{3}{4})$. Student's answer: $\frac{1}{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The product of a negative and a positive is negative. The answer is $-\frac{6}{12} = -\frac{1}{2}$.**

Correct answer: **$-\frac{1}{2}$**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A team gains 3 points, then loses 5 points. What is the net change? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The loss is negative. 3 + (-5) = -2 points.**

Correct answer: **-2 points**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

The temperature falls from −4°F to −12°F. What is the change in temperature? Student's answer: 8°F

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The temperature went down, so the change is negative. −12 − (−4) = −8°F.**

Correct answer: −8°F

Find the Error

Grade 7 · 7.NS.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A fish sinks 2.5 feet every minute for 4 minutes. What is its position change in feet? Student's answer: 10 feet

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Sinking is a negative change. 2.5 × 4 = 10, so the change is -10 feet.**

Correct answer: **-10 feet**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $3(x + 2) - x$. Student writes: $2x + 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 must multiply the 2 as well: 3x + 6. Then 3x + 6 − x = 2x + 6.**

Correct answer: **2x + 6**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify 2x + 5 − 4x + 1. Student writes: −2x + 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The constant terms are 5 + 1 = 6, not 4. The answer is −2x + 6.**

Correct answer: **−2x + 6**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $4(2y - 1) + 3$. Student writes: $8y + 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **$4 \times (-1) = -4$, not $+4$ or 0. So $8y - 4 + 3 = 8y - 1$.**

Correct answer: **$8y - 1$**

Find the Error

Grade 7 · 7.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 4x + 10 in factored form. Student writes: 4(x + 2.5)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The factor 2.5 is not an integer. The GCF of 4 and 10 is 2, so the answer is 2(2x + 5).**

Correct answer: **2(2x + 5)**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An item is 15% off a price p. Which expression shows the sale price? Student writes: 0.15p

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **0.15p is the discount. The price after 15% off is 0.85p.**

Correct answer: **0.85p**

Find the Error

Grade 7 · 7.EE.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Factor 9y - 6. Student writes: 3(3y + 2)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The sign of the 2 is wrong. 3(3y + 2) = 9y + 6. The correct factored form is 3(3y - 2).**

Correct answer: **3(3y - 2)**

Find the Error

Grade 7 · 7.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Estimate 4.8 × 6.2. Student's estimate: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Round 4.8 to 5 and 6.2 to 6. 5 × 6 = 30, so the estimate is about 30.**

Correct answer: **About 30**

Find the Error

Grade 7 · 7.EE.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Four notebooks cost $6.75 each, plus $3 for tax. What is the total? Student's answer: $24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted the tax. 4 × $6.75 = $27, and $27 + $3 = $30.**

Correct answer: **$30**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 2x + 3 = 11. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student stopped after subtracting 3. Divide by 2: x = 4.**

Correct answer: **4**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 4x + 5 = 25. Student's answer: 7.5

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 5 to 25. Subtract 5: 4x = 20, so x = 5.**

Correct answer: **5**

Find the Error

Grade 7 · 7.EE.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $3(x + 1) = 18$. Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **18 ÷ 3 = 6, so x + 1 = 6. Subtract 1 to get x = 5.**

Correct answer: **5**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve 2x - 7 = 9. Student's answer: 16

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found 2x = 16 and stopped. Divide by 2: x = 8.**

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A map scale is 1 cm = 5 km. Two cities are 3.5 cm apart on the map. How far apart are they?

Student's answer: 3.5 km

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not use the scale. 3.5 × 5 = 17.5 km.**

Correct answer: **17.5 km**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A drawing has a scale of 1 inch = 4 feet. A room is 12 feet long. How long is it in the drawing?

Student's answer: 48 inches

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 12 by 4. Divide: 12 ÷ 4 = 3 inches.**

Correct answer: **3 inches**

Find the Error

Grade 7 · 7.G.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A scale drawing uses 1 cm = 2 m. A wall is 6 m long. How long is it in the drawing? Student's answer: 12 cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied instead of dividing. 6 ÷ 2 = 3 cm.**

Correct answer: **3 cm**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have sides of 3 cm, 4 cm, and 5 cm? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **3 + 4 is greater than 5, so these sides make a triangle. It is a right triangle.**

Correct answer: **Yes**

Find the Error

Grade 7 · 7.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have sides of 2 cm, 3 cm, and 6 cm? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **2 + 3 = 5, which is less than 6. These sides cannot form a triangle.**

Correct answer: **No**

Find the Error

Grade 7 · 7.G.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Can a triangle have angles of 60°, 60°, and 60°? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **60 + 60 + 60 = 180, so the angles can form a triangle. It is an equilateral triangle.**

Correct answer: **Yes**

Find the Error

Grade 7 · 7.G.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts a cube parallel to one of its faces. What is the shape of the cross-section?

Student's answer: triangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A slice parallel to a face has the same shape as that face. The cross-section is a square.**

Correct answer: **Square**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts a cylinder parallel to its bases. What is the shape of the cross-section? Student's answer: rectangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A slice parallel to the base has the same shape as the base. The cross-section is a circle.**

Correct answer: **Circle**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A plane cuts through two opposite edges of a cube. What is the shape of the cross-section?

Student's answer: triangle

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The slice passes through two opposite edges and crosses the cube as a rectangle.**

Correct answer: **Rectangle**

Find the Error

Grade 7 · 7.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the circumference of a circle with radius 3 cm. Student's answer: 9π cm

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied 3 by 3. $C = 2\pi \times 3 = 6\pi$, which is about 18.85 cm.**

Correct answer: **6π cm, about 18.85 cm**

Find the Error

Grade 7 · 7.G.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a circle with diameter 10 cm. Student's answer: 100π cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the diameter in the area formula. The radius is 5, so $A = \pi \times 5^2 = 25\pi$ cm².**

Correct answer: **25π cm², about 78.5 cm²**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a circle with radius 5 cm. Student's answer: 10π cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student doubled the radius. $A = \pi \times 5^2 = 25\pi$ cm².**

Correct answer: **25π cm²**

Find the Error

Grade 7 · 7.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are supplementary. One is 65°. What is the other? Student's answer: 25°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 90°, which is for complementary angles. 180 - 65 = 115°.**

Correct answer: **115°**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are complementary. One is 35°. What is the other? Student's answer: 145°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 180°. 90 - 35 = 55°.**

Correct answer: **55°**

Find the Error

Grade 7 · 7.G.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two angles are vertical angles. One is 70°. What is the other? Student's answer: 110°

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Vertical angles are equal. The other angle is 70°.**

Correct answer: **70°**

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A triangle has a base of 8 cm and a height of 5 cm. What is its area? Student's answer: 40 cm²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot the $\frac{1}{2}$. A = $\frac{1}{2}$ × 8 × 5 = 20 cm².**

Correct answer: **20 cm²**

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A circular table has a radius of 3 feet. What is its area, to the nearest hundredth? Student's answer: 18.85 ft²

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **18.85 is the circumference. The area is π × 3² = 9π, which is about 28.27 ft².**

Correct answer: **About 28.27 ft²**

Find the Error

Grade 7 · 7.G.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A prism has a base area of 12 square units and a height of 5 units. What is its volume? Student's answer: 17 cubic units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 12 and 5. Multiply: 12 × 5 = 60 cubic units.**

Correct answer: **60 cubic units**

Find the Error

Grade 7 · 7.SP.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coach asks 20 players on the basketball team their favorite sport to learn about the whole school. Is this sample representative? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The sample is only basketball players, so it is not representative of the whole school.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 50 students is taken from a school of 500. Is the sample likely to be representative? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A random sample is usually representative of the population, so this sample is likely representative.**

Correct answer: **Yes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A survey asks the first 15 people in line at an ice cream shop about flavors for all the students. Is this sample representative? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The sample is not random, and it includes only people at the shop. It is not representative of all students.**

Correct answer: **No**

Find the Error

Grade 7 · 7.SP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a random sample of 50 students from a school of 800, 20 students said they like the new lunch menu. Estimate how many students in the school like it. Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the sample count. $\frac{20}{50}$ = 40%, and 40% of 800 = 320.**

Correct answer: 320

Find the Error

Grade 7 · 7.SP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In a random sample of 100 students, 60 said yes to a question. Estimate the number of yes answers in a population of 1,500. Student's answer: 60

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student repeated the sample count. $\frac{60}{100}$ = 60%, and 60% of 1,500 = 900.**

Correct answer: **900**

Find the Error

Grade 7 · 7.SP.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 25 students has 8 who like soccer. What is the estimated percent of the population who like soccer? Student's answer: 8%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **8 out of 25 is 32%, not 8%.**

Correct answer: **32%**

Find the Error

Grade 7 · 7.SP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has data 10, 12, 14. Group B has data 11, 13, 15. Which group has the higher mean?

Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Group A's mean is 12, and Group B's mean is 13. Group B has the higher mean.**

Correct answer: **Group B**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a range of 3 and Group B has a range of 10. Which group is more variable?

Student's answer: Group A

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A larger range means more variability. Group B is more variable.**

Correct answer: **Group B**

Find the Error

Grade 7 · 7.SP.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group A has a median of 22 and Group B has a median of 33. What is the difference between the medians? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted the wrong numbers. 33 - 22 = 11.**

Correct answer: **11**

Find the Error

Grade 7 · 7.SP.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Group X has a mean of 12 and a mean absolute deviation of 2. Group Y has a mean of 15 and a mean absolute deviation of 2. Is the difference in means large compared to the variability?
Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The means differ by 3, which is larger than the spread of 2. The difference is meaningful.**

Correct answer: **Yes**

Find the Error

Grade 7 · 7.SP.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A random sample of 30 students from School A has a mean of 7 hours of sleep. A random sample from School B has a mean of 7 hours. Student says: The schools have different sleep amounts.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The means are the same, so there is no difference to infer from these samples.**

Correct answer: **No difference shown**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two random samples have medians of 40 and 42, with spread of 15 in each. Student says: The difference is meaningful.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A difference of 2 is small compared to a spread of 15. The difference is not meaningful.**

Correct answer: **Not meaningful**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A weather app says the chance of rain is 1.5. Can this be correct? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Probabilities must be between 0 and 1. A value of 1.5 is not possible.**

Correct answer: **No**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the probability of an impossible event? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **An impossible event has probability 0. An event that is certain has probability 1.**

Correct answer: 0

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the probability of rolling a number less than 7 on a standard die? Student's answer: 0

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Every roll is less than 7, so the event is certain. Its probability is 1.**

Correct answer: 1

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin is flipped 50 times and lands heads 28 times. What is the experimental probability of heads? Student's answer: 0.28

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not divide by the total. $\frac{28}{50}$ = 0.56.**

Correct answer: 0.56

Find the Error

Grade 7 · 7.SP.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A spinner is spun 100 times and lands on red 20 times. What is the experimental probability of red? Student's answer: 20

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave the count. The probability is $\frac{20}{100}$ = 0.2.**

Correct answer: **0.2**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin has a theoretical probability of heads of 0.5. A student flips it 10 times and gets 3 heads. Student says: The coin must be broken.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Experimental results vary, especially with few trials. 10 flips is too few to say the coin is broken.**

Correct answer: **No, the coin is not necessarily broken**

Find the Error

Grade 7 · 7.SP.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A fair six-sided die is rolled. What is the probability of rolling a 4? Student's answer: $\frac{1}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 6 equally likely outcomes, so P(4) = $\frac{1}{6}$.**

Correct answer: $\frac{1}{6}$

Find the Error

Grade 7 · 7.SP.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model gives outcomes A, B, C, and D with probabilities 0.2, 0.3, and 0.4 for A, B, and C. What is P(D)? Student's answer: 0.2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **0.2 + 0.3 + 0.4 = 0.9. P(D) = 1 − 0.9 = 0.1.**

Correct answer: **0.1**

Find the Error

Grade 7 · 7.SP.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A spinner has 8 equal sections, and 3 are blue. What is P(blue)? Student's answer: $\frac{3}{5}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The total is 8 sections, not 5. P(blue) = $\frac{3}{8}$.**

Correct answer: **$\frac{3}{8}$**

Find the Error

Grade 7 · 7.SP.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two fair coins are flipped. What is the probability of getting two heads? Student's answer: $\frac{1}{2}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The outcomes are HH, HT, TH, and TT. Only HH works, so P = $\frac{1}{4}$.**

Correct answer: $\frac{1}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Two fair dice are rolled. What is the probability that the sum is 7? Student's answer: $\frac{1}{12}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **There are 6 pairs that make 7 out of 36 total outcomes. $\frac{6}{36} = \frac{1}{6}$.**

Correct answer: $\frac{1}{6}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A coin is flipped and a die is rolled. How many total outcomes are there? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The coin has 2 outcomes and the die has 6. The total is 2 × 6 = 12.**

Correct answer: **12**